RES POLYTECHNIC, BANGALORE-91

CIE Theory Test III - Model Answers

System and Network Administration | Course Code 25CS34I | Marks 50 | Semester III

SECTION I

Q1 a) Bootable pen drive and installing Windows 10/11 (5 marks)

1. Take a pen drive (8 GB or more) and back up its data (it will be formatted). Download the Windows ISO / Media Creation Tool from Microsoft.
2. Insert the pen drive and run Rufus (or Media Creation Tool). Select the drive and the ISO, choose partition scheme (GPT for UEFI / MBR for legacy BIOS), click **Start**.
3. Insert the bootable pen drive in the target computer, restart and press the boot-menu key (F12 / F2 / Esc / Del) and select the USB drive.
4. Windows Setup starts: choose language, click **Install now**, enter product key (or "I don't have a key").
5. Select **Custom install**, choose/create the partition (delete old partitions if needed) and click Next.
6. Files are copied and the PC restarts several times; remove the pen drive when asked.
7. Complete setup: region, user account, password, privacy settings. Install drivers and updates.

Q1 b) "Automatic Repair" loop after a power cut (10 marks)

Cause: sudden power cut has damaged system files / boot records / the registry, so Windows cannot start normally.

Identify: on the repair screen choose **Advanced options → Troubleshoot → Advanced options**. Check the error in *Startup Settings* and *Command Prompt* (`chkdsk C: /f`, `sfc /scannow`, `bootrec /fixmbr`, `bootrec /fixboot`, `bootrec /rebuildbcd`).

Option	What it does	When to choose
Safe Mode	Starts Windows with minimum drivers and services	First option, when the problem is a faulty driver/software/update; remove or roll back the recent change
System Restore	Returns system files, registry and settings to an earlier restore point (personal files are safe)	When a restore point exists from before the problem and the PC fails after a recent change
System Reset (Reset this PC)	Reinstalls Windows; can keep files or remove everything	When restore and repair fail, no restore point or image exists, and system files are badly corrupted
Image Recovery	Restores the full system from a previously created system image backup	When a full backup image exists; used for disk failure or total corruption (restores everything, but later data is lost)

Order of steps: Startup Repair → Safe Mode → System Restore → Image Recovery → Reset this PC (last resort, as it is the most destructive).

Q1 c) User account and folder protection commands (10 marks)

Command Prompt (run as Administrator):

```
net user CSE_LAB1_02 Lab@1234 /add
net user CSE_LAB1_02 /fullname:"CSE Lab1 Student 02" /passwordchg:no
net user CSE_LAB1_02 /active:no
net user CSE_LAB1_02 /delete
cipher /e /s:D:\Records
cipher /d /s:D:\Records
```

Command	Explanation
`net user CSE_LAB1_02 Lab@1234 /add`	Adds a new local user with the given password
`net user CSE_LAB1_02 /fullname:"..." /passwordchg:no`	Modifies the account (full name; user cannot change password). `/active:no` disables the account
`net user CSE_LAB1_02 /delete`	Deletes the user account
`cipher /e /s:D:\Records`	Encrypts the folder and its sub-folders (EFS)

`cipher /d /s:D:\Records`	Decrypts the folder and its sub-folders

PowerShell equivalents:

```
New-LocalUser -Name "CSE_LAB1_02" -Password (ConvertTo-SecureString "Lab@1234" -AsPlainText -Force)
Set-LocalUser -Name "CSE_LAB1_02" -FullName "CSE Lab1 Student 02"
Remove-LocalUser -Name "CSE_LAB1_02"
```

Note: EFS encryption needs an NTFS drive and Windows Pro/Enterprise; only the encrypting user can open the files.

OR

Q2 a) FAT32 vs NTFS vs exFAT (5 marks)

Feature	FAT32	NTFS	exFAT
Max file size	4 GB	Very large (16 TB+)	16 EB (practically unlimited)
Max volume size	32 GB (Windows format) / 2 TB	256 TB+	Very large
Security	No permissions or encryption	File permissions, EFS encryption, quotas	No permissions or encryption
Journaling / recovery	No	Yes (journaling)	No
Best used for	Small pen drives, old devices	Windows system drive	Large pen drives, SD cards, external drives
Compatibility	Very high (all OS)	Windows (read-only on macOS)	Windows, macOS, Linux, cameras

Q2 b) Slow startup and slowdown after one hour (10 marks)

Finding the cause:

- **Task Manager (Ctrl+Shift+Esc):** *Processes* tab shows apps using high CPU, memory or disk; *Startup* tab shows startup impact; *Performance* tab shows CPU/RAM/Disk/Network graphs. If memory keeps growing, a memory leak is likely.
- **Resource Monitor (resmon):** detailed CPU, memory, disk and network usage per process; shows which process is causing 100% disk usage or heavy network usage.
- **Event Viewer (eventvwr):** *Windows Logs → System/Application* – look for Critical/Error/Warning events (disk errors, service failures, driver crashes); *Diagnostics-Performance* shows boot time delays.

Improving performance:

1. Disable unnecessary **startup applications** (Task Manager → Startup).
2. Stop/disable unwanted **services** (`services.msc`), set non-essential ones to Manual.
3. Run **Disk Cleanup** (temp files, recycle bin, update cache) and remove unused programs.
4. Scan for **malware** (Windows Security full scan).
5. **Update drivers** and Windows (Device Manager).
6. Check **disk health** (`chkdsk C: /f`, SMART status); defragment HDD / TRIM SSD; replace a failing disk.
7. Add RAM / upgrade to SSD if hardware is the bottleneck.

Q2 c) 3 TB disk showing 2 TB; MBR vs GPT; partitions (10 marks)

Reason: the disk is initialised as **MBR**, which supports a maximum of 2 TB per disk and only 4 primary partitions. Converting to **GPT** allows the full 3 TB.

MBR	GPT
Master Boot Record	GUID Partition Table
Max 2 TB disk	Up to 9.4 ZB (full 3 TB usable)
Max 4 primary partitions	Up to 128 partitions
Legacy BIOS boot	UEFI boot
One copy of partition table (less reliable)	Backup copy and CRC check (more reliable)

Steps in Disk Management (`diskmgmt.msc`):

1. Right-click the disk → check **Properties → Volumes** for partition style (MBR/GPT). If it is MBR and empty, right-click the disk → **Convert to GPT Disk** (back up data first; the existing volumes must be deleted).

2. Right-click the unallocated space → **New Simple Volume** → set size (e.g. 1 TB) → assign drive letter → select file system (NTFS) and label → *Format* → Finish. Repeat for the three drives.
3. **Shrink:** right-click a volume → **Shrink Volume** → enter amount in MB → Shrink (creates unallocated space).
4. **Extend:** right-click a volume → **Extend Volume** (needs adjacent unallocated space) → Next → Finish.

SECTION II

Q3 a) VPN and two types (5 marks)

VPN (Virtual Private Network): creates an encrypted, secure tunnel over a public network (the Internet) between a user/site and a private network, hiding the IP address and protecting data from eavesdropping.

Types (any two):

1. **Remote-access VPN:** an individual user connects from home or travel to the office network using VPN client software (e.g. staff working from home).
2. **Site-to-Site VPN:** connects two entire networks (e.g. head office and branch office) through VPN gateways/routers; users need no client software.
3. **Client-based/SSL VPN:** browser or app-based access (also used in remote access).

Q3 b) Website opens by IP but not by name (10 marks)

Problem: a **DNS (name resolution) problem** – the network connection works, but the name is not converted into an IP address (DNS server wrong/not reachable or corrupted DNS cache).

Diagnosis and solution:

1. `ipconfig /all` – check IP, gateway and **DNS server** addresses.
2. `ping 8.8.8.8` (or the site IP) – replies show the Internet connection is working.
3. `ping www.example.com` – "could not find host" confirms DNS failure.
4. `tracert www.example.com` / `nslookup www.example.com` – check where the path or lookup fails.
5. `ipconfig /flushdns` – clear the corrupted DNS cache (also `ipconfig /registerdns`).
6. Change DNS server (Adapter → IPv4 Properties) to a working one, e.g. 8.8.8.8 / 1.1.1.1, or the router's address. Test again.

Roles:

- **DNS server:** converts domain names to IP addresses (like a phone book).
- **Default gateway:** the router that forwards packets from the local network to other networks/Internet; it is needed to reach the website once the IP is known.

Q3 c) Folder sharing, firewall, network profile, secure remote access (10 marks)

1. Set the network profile: Settings → Network → Properties → choose **Private** (trusted office network; discovery and sharing on). Use **Public** on unknown networks (sharing blocked).

2. Share the folder: right-click folder → Properties → **Sharing → Advanced Sharing** → tick *Share this folder* → **Permissions**: remove *Everyone*, add only the staff group/users (Read or Change). Also set **Security (NTFS) permissions** for the same users. Staff access it by `\\ComputerName\ProjectFolder`.

3. Windows Firewall: Control Panel → Windows Defender Firewall → keep it **ON** for Private and Public profiles; allow *File and Printer Sharing* only for the **Private** profile; block inbound connections on Public; create inbound rules to allow only the office subnet; unused ports stay blocked. This keeps outsiders out.

4. Secure remote access for working from home:

- Set up a **VPN** to the office (remote-access VPN).
- After the VPN connects, use **Remote Desktop (RDP)** or access the shared folder; do not expose RDP directly to the Internet.
- Use strong passwords, Multi-Factor Authentication, Network Level Authentication, keep Windows/antivirus updated, and limit access to named users.

OR

Q4 a) Virtual machine (hypervisor) (5 marks)

A **virtual machine** is a software-based computer that runs its own operating system inside the host computer using virtual CPU, RAM, disk and network. A **hypervisor** is the software that creates and manages VMs (e.g. Hyper-V, VirtualBox, VMware Workstation).

Benefits for testing a new OS (any two):

1. **Safe testing:** the host system is not affected if the new OS crashes or has malware; VM is isolated.
2. **No extra hardware / no partitioning:** several OS can run on one computer at the same time.
3. Snapshots allow instant rollback to a previous state; VMs can be copied, cloned and deleted easily; low cost.

Q4 b) E-waste disposal of 30 computers (10 marks)

E-waste: discarded, obsolete or damaged electrical and electronic equipment. **Examples:** old computers, monitors, keyboards, printers, batteries, mobile phones, UPS, cables, circuit boards.

Disposal using the e-waste lifecycle:

1. **Collection:** make an inventory of the 30 computers (asset register); back up and permanently erase data (wipe/format/destroy drives); store in a designated area; hand over only to an **authorised collection centre/recycler** (not scrap dealers). Obtain a receipt/certificate.
2. **Sorting:** separate into reusable/repairable items (donate or refurbish), recyclable parts (metals, plastic, glass, batteries) and hazardous materials (CRT glass, batteries, lead, mercury).
3. **Recovery:** the authorised recycler dismantles the items and recovers valuable materials (copper, gold, aluminium, plastics); hazardous parts are treated and disposed of safely; the remainder goes to approved disposal.

Impact of improper disposal:

- **Environment:** toxic metals (lead, mercury, cadmium) leach into soil and water; burning releases toxic gases and pollutes air; landfills fill up.
- **Health:** exposure causes kidney, lung and nervous-system damage, cancer, and affects children and informal recycling workers.

Websites for latest rules: Central Pollution Control Board (**cpcb.nic.in**), Ministry of Environment, Forest and Climate Change (**moef.gov.in**), and the E-Waste (Management) Rules, 2022 as updated.

Q4 c) Computer lab setup plan for 30 students (10 marks)

Area	Plan
1. Hardware purchase	30 student PCs + 1 faculty PC (i5/Ryzen 5, 8–16 GB RAM, SSD, monitor); 1 server; 24/48-port managed switch, wireless access point, router/firewall; printer/projector; UPS with battery backup (e.g. 30 × 600 VA or central UPS); furniture. Prepare specification, get 3 quotations, compare cost/warranty/after-sales support, issue purchase order, check delivery against the order.
2. Installation	Lab layout with proper spacing and ventilation (AC/fans); structured LAN cabling (Cat6, labelled, cable trays); install OS using an image/clone, drivers, required software and antivirus; configure IP/DHCP, DNS, user accounts and shared folders; test every system and the network; maintain an asset register and stock record.
3. Electrical safety	Proper earthing of all equipment; MCB/ELCB (RCCB) and surge protection; separate circuits for computers and AC; good-quality wiring; no overloaded sockets; UPS for backup; fire extinguisher (CO_2) and smoke detector; first-aid kit; keep liquids away; emergency switch; periodic inspection.
4. Cybersecurity	Firewall; antivirus and regular updates/patches; separate user accounts with limited rights and strong passwords; disable USB if not needed; content filtering and logging; regular data backups; guest/student Wi-Fi separated by VLAN; physical security (lock, CCTV, access register); awareness about phishing.